# Whamboard Prototype

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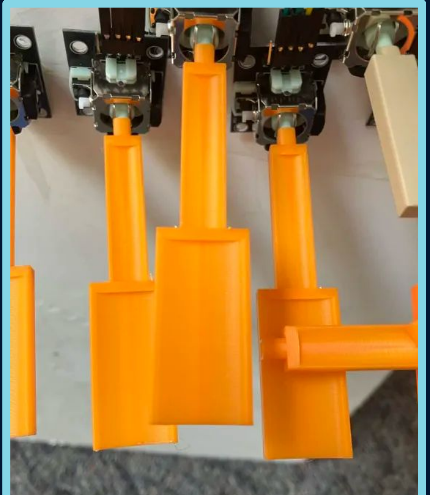
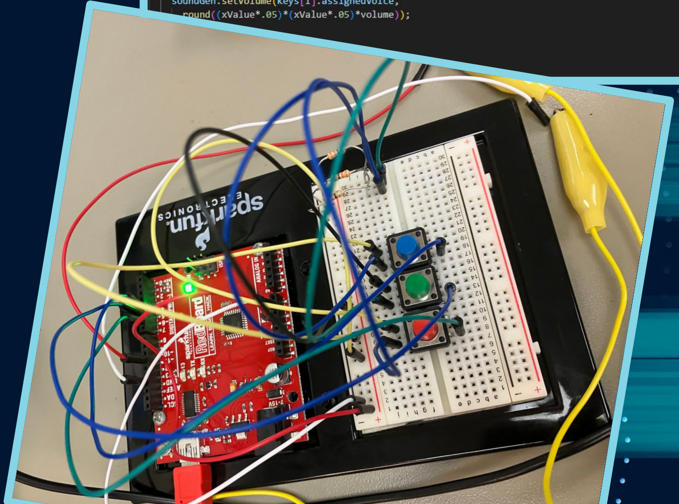
# Individual Contributions – Kavan

- Input programming
- *Wavetable generation
- Modeling and 3D printing piano keys
- Wiring…

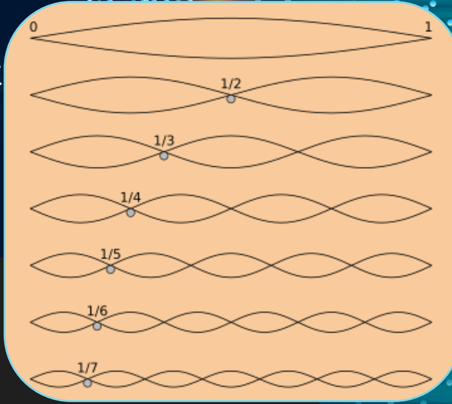
```
for(int i = 0; i < numOfKeys; i++){

  float xValue = (xValues[i]-500)-inputBuffer;
  float yValue = (yValues[i]-500)-inputBuffer;

  if(keys[i].pressed == false && xValue > 0){
    Serial.print("key down: ");
    Serial.print(i);
    Serial.println();
    //int voice = getVoiceNumber();
    int voice = i; //temporary fix for weirdness! come back and fix this!!!
    if(voice != -1){
      //soundGen.mTrigger(voice, round(baseNote+7-i*3.33));
      keys[i].pressed = true;
      soundGen.setFrequency(voice, noteFrequency(round(baseNote-i)));
      soundGen.setVolume(voice, round((xValue*.05)*(xValue*.05)*volume));
      soundGen.trigger(voice);
      keys[i].assignedVoice = voice;
    }
  }
  if(keys[i].pressed == true && abs(xValues[i]-500) < inputBuffer){
    Serial.print("key up: ");
    Serial.print(i);
    Serial.println();
    keys[i].pressed = false;
    soundGen.release(keys[i].assignedVoice);
  }

  if(keys[i].pressed){

    //note bend
    if(abs(yValue) > 20){
      float bendVal = yValue - 20 * signum(yValue);
      bendVal *= -bendPower;
      soundGen.setFrequency(keys[i].assignedVoice, noteFrequency(round(baseNote-i)+bendVal));
    }else{
      soundGen.setFrequency(keys[i].assignedVoice, noteFrequency(round(baseNote-i)));
    }

    //joystick modular volume
    soundGen.setVolume(keys[i].assignedVoice,
      round((xValue*.05)*(xValue*.05)*volume));
```





# Wavetable Generation from Overtones

- Wavetable = numbers that can be played over and over = instrument
- Overtones = sounds of instruments separated into sinewaves
- Our wavetable generator takes overtones and makes wavetables

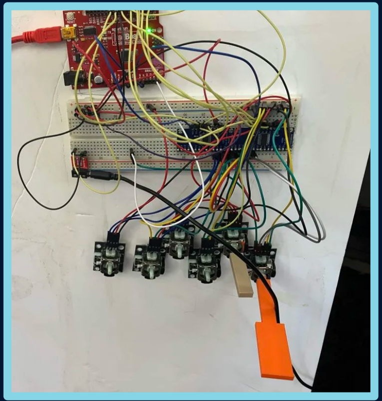



```python
import math

overtones = [1, .1, .5, .1, .25, .1, .125, .05]

overtone_total = 0
for i in range(0, len(overtones)):
    overtone_total += overtones[i]

with open("wavetable.txt", "w") as f:
    for i in range(0, 256):
        output = 0
        for j in range(0, len(overtones)):
            output += 126 * (overtones[j]/overtone_total) * math.sin((i * 2 * math.pi * (1/256) * (j+1) ))

        if output > 126:
            output = 126
        if output < -126:
            output = -126

        print( str(round(output)) + ",\t//" + str(i))
        print( str(round(output)) + ",\t//" + str(i), file = f)
```

```
0,  //0
7,  //1
13, //2
20, //3
26, //4
32, //5
38, //6
44, //7
49, //8
55, //9
59, //10
64, //11
68, //12
71, //13
75, //14
77, //15
80, //16
82, //17
83, //18
84, //19
85, //20
85, //21
85, //22
84, //23
83, //24
```

# Individual Contributions – Weston

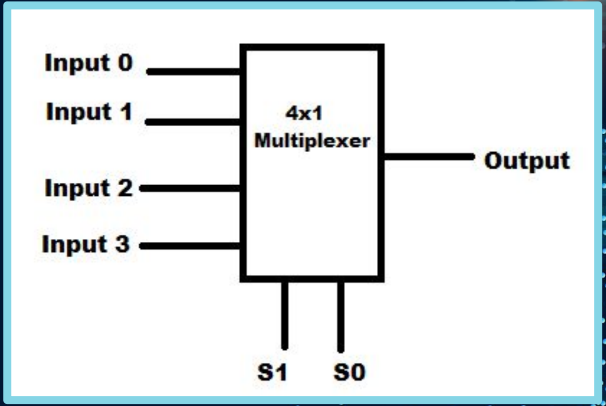
- *Soldering MUX boards
- Preset function
- Physical prototyping
- *Relative pitch for tuning



```
    //Preset function
    //cycles through different instruments (simple waves and evelopes fo
    if (digitalRead(buttonPin) == HIGH && false){//disabled for now
      if(!specialButtonDown){
        //Serial.print(preset);
        if(preset == 1){

          for(int i = 0; i < 4; i++){
            soundGen.setupVoice(i, CUSTOMA, 60, ENVELOPE0, 90, 64);
          }

        }else if(preset == 2){

          for(int i = 0; i < 4; i++){
            soundGen.setupVoice(i, CUSTOMB, 60, ENVELOPE1, 90, 64);
          }

        } else {
            for(int i = 0; i < 4; i++){
            soundGen.setupVoice(i, CUSTOMC, 60, ENVELOPE2, 90, 64);
            }
            //the next time the button is pressed, it will reset to 0 an
            preset = 0;
        }
        preset = preset += 1;
      }
      specialButtonDown = true;
    }else{
      specialButtonDown = false;
    }

  }
```

# Problem Solving / Struggles

- Modeling ergonomic & playable keys
- Making functional keys
- Tuning
- Adding new effects
- *Wiring
- *Using MUX boards



# Lesson

Wires are hard to organize, and confusing wiring has spawned many issues during this project. Going into the next term, we'll focus more on wiring our board in a neat and justified fashion.

# Plan

## Almost there?

kinda?

### Required

- An octave (12 keys)
- Sound settings

### Nice to Have

- Two octaves (24 keys)
- MIDI Input / Output
- 3D printed keyboard case (not just breadboard)

### Superfluous

- More octaves
- LCD display